

Python Code

```
1  import pandas as pd
2  import matplotlib.pyplot as plt
3
4  data = {
5      'department': ['IT', 'CSE', 'ECE', 'Civil'],
6      'students': [50, 70, 40, 30]
7  }
8
9  df = pd.DataFrame(data)
10
11 print(df)
12
13 df.plot(
14     kind='pie',
15     y='students',
16     labels=df['department'],
17     autopct='%1.1f%%',
18     title='Department Students'
19 )
20
21 plt.show()
22
23
24 import pandas as pd
25 import matplotlib.pyplot as plt
26
27 data = {
28     'month': ['Jan', 'Feb', 'Mar', 'Apr'],
29     'sales': [100, 150, 120, 180]
30 }
31
32 df = pd.DataFrame(data)
33
34 print(df)
35
36 df.plot(
37     kind='area',
38     x='month',
39     y='sales',
40     title='Monthly Sales'
41 )
42
43 import pandas as pd
44 import matplotlib.pyplot as plt
45
46 data = {
47     'marks': [45, 55, 60, 70, 80, 90]
48 }
49
50 df = pd.DataFrame(data)
```

```
51
52 print(df)
53
54 df.plot(
55     kind='box',
56     title='Marks Analysis'
57 )
58
59 plt.show()
60
61 import pandas as pd
62 import matplotlib.pyplot as plt
63
64 data = {
65     'course': ['Python', 'AI', 'Java', 'SQL'],
66     'students': [40, 60, 35, 50]
67 }
68
69 df = pd.DataFrame(data)
70
71 print(df)
72
73 df.plot(
74     kind='barh',
75     x='course',
76     y='students',
77     title='Course Enrollment'
78 )
79
80 plt.show()
81
82
```

Figure 1

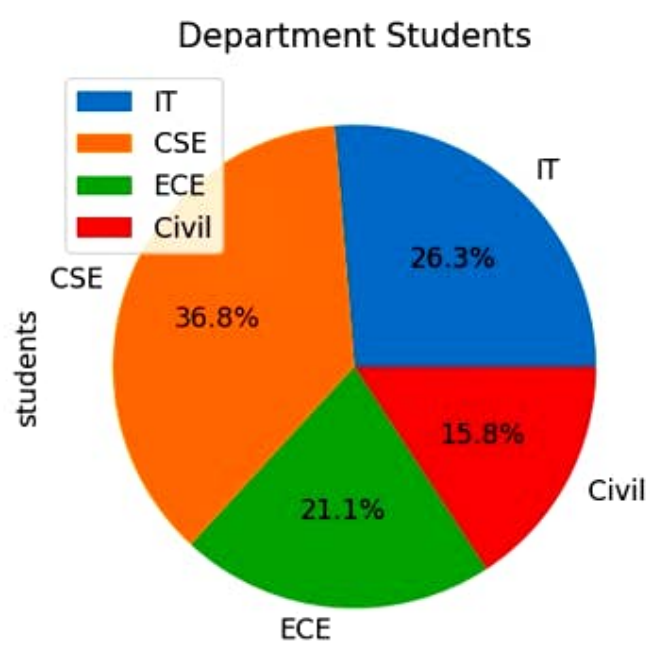
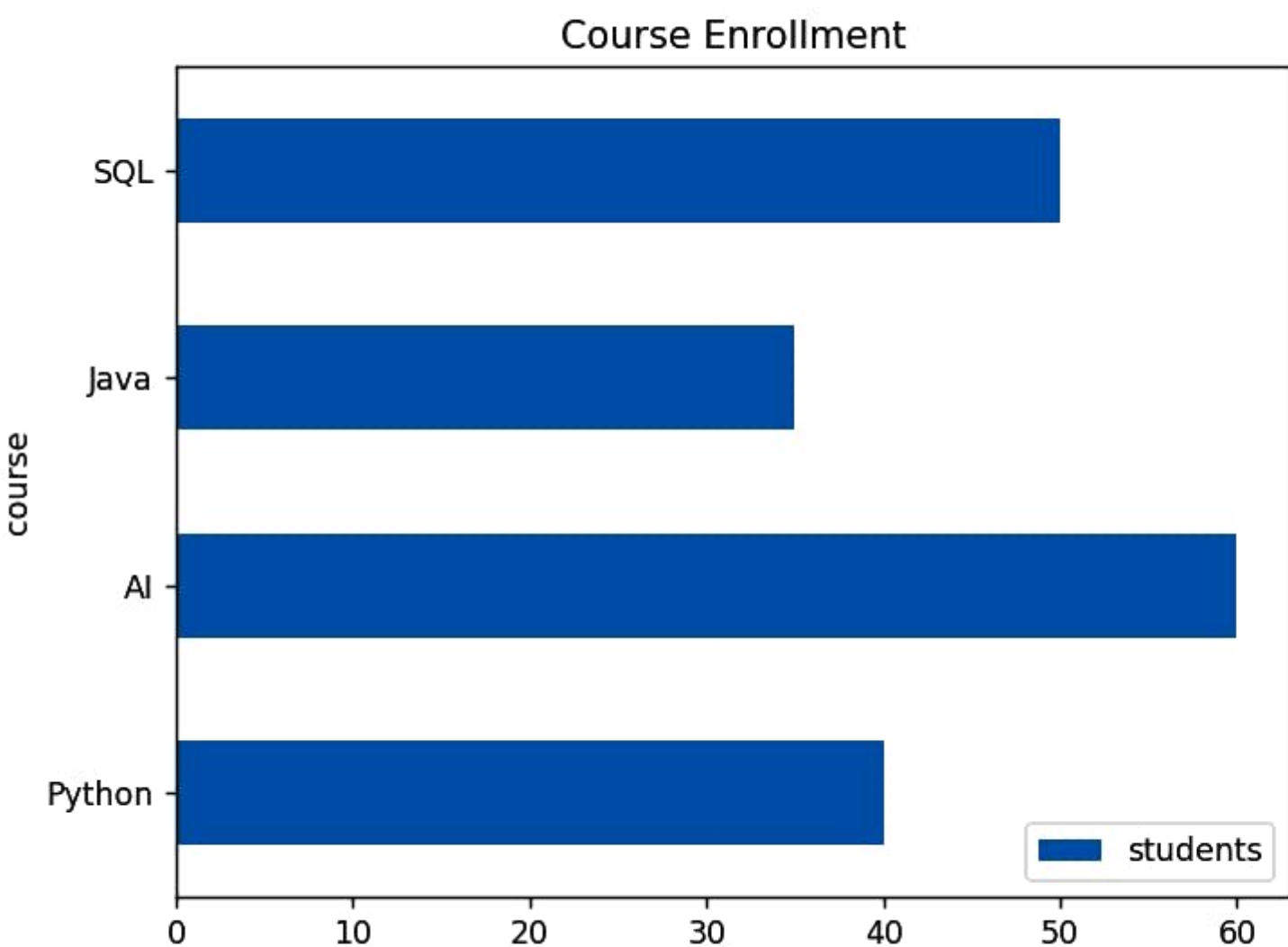


Figure 1

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(x, y) = (12.4,)

Figure 2

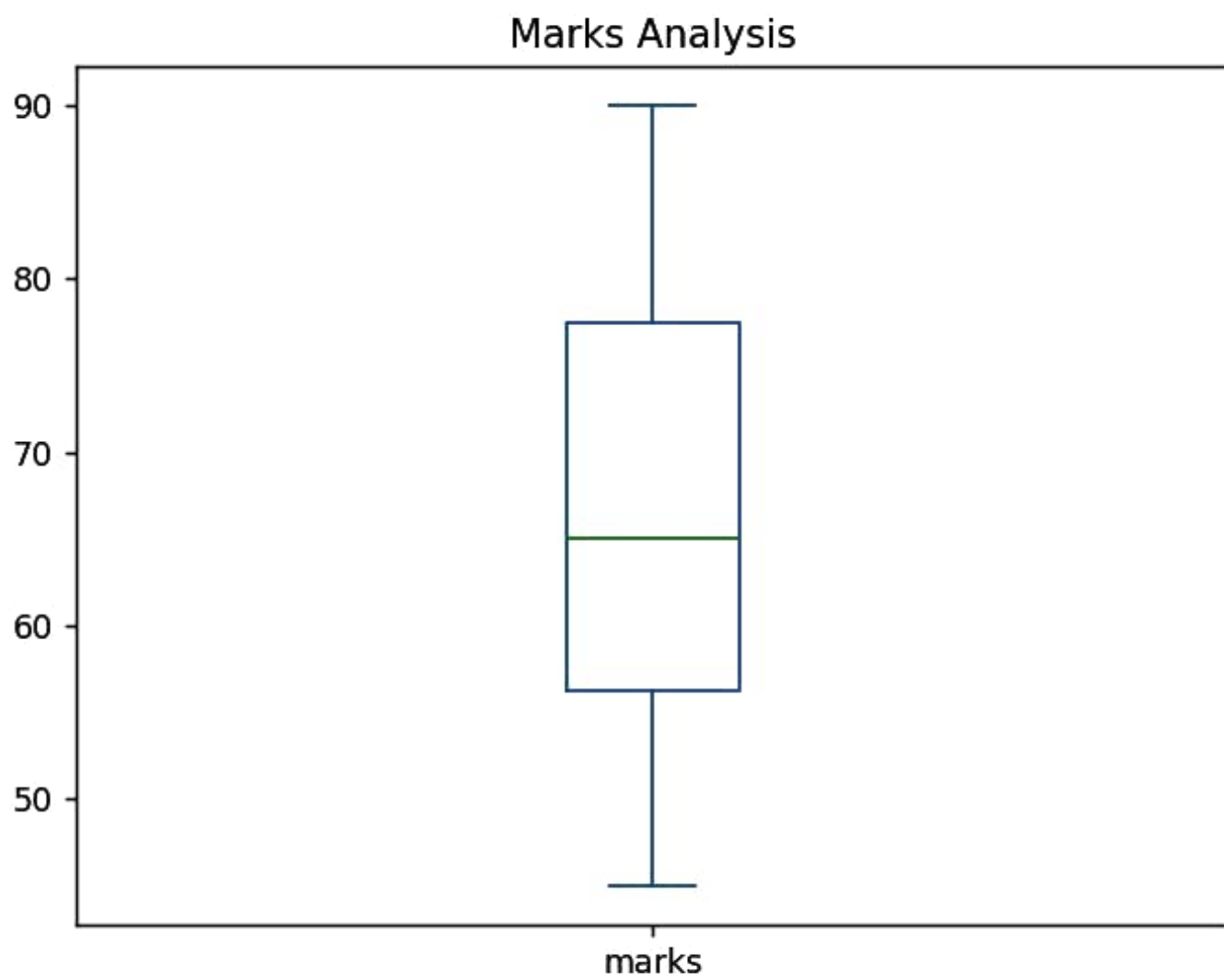




Figure 1



Monthly Sales

